The Checkpoint Selection Problem Is an Evaluation Budget Problem: A $\Delta_{\min}$ Framework and Six Analytical Traps

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Abstract

Given ten checkpoints from an LLM fine-tuning run, how confidently can you pick the best one? The standard answer is to pick the checkpoint with the lowest validation loss. This is known to be imperfect, but the nature of its imperfection is poorly characterized. We investigate this question through a controlled LoRA fine-tuning of Qwen2.5-7B (500 steps, 10 checkpoints, multi-task data) and find that the evaluation noise floor itself (the minimum quality difference detectable with a given held-out set) can exceed the actual differences between checkpoints. At $N = 200$ held-out samples, the minimum detectable difference is 9.8 percentage points; the observed range across all checkpoints is 8.0 percentage points on a mathematical reasoning task, and 6.4 pp on a knowledge task, and no checkpoint-to-checkpoint difference reaches statistical significance ($\min p = 0.161$). Along the way, we encountered and document six analytical traps that can produce false conclusions about proxy-quality misalignment. These traps, the $\Delta_{\min}$ framework, and the frozen-definitions methodology we developed are the paper's primary contribution: a diagnostic checklist for practitioners to apply before investing in checkpoint selection strategies.

1 Introduction

Given ten checkpoints from an LLM fine-tuning run, how confidently can you pick the best one? The standard answer is to pick the checkpoint with the lowest validation loss. This approach is known to be imperfect. But the nature of its imperfection is poorly understood: is it a rare failure mode triggered by distribution shift or overfitting, or is it a systematic limitation of the practice itself?

This paper reports on a diagnostic investigation. We set up a standard LoRA fine-tuning of Qwen2.5-7B (500 steps, multi-task data), generated 10 checkpoints, and attempted to answer three questions:

1. Given a held-out evaluation set of size N, what is the minimum quality difference between checkpoints that can be reliably detected?
2. In this fine-tuning run, how do actual checkpoint differences compare to this threshold?
3. What analytical traps can lead to false conclusions about proxy-quality misalignment?

The answers, in brief:

1. At $N = 200$ (a typical evaluation budget), the minimum detectable difference is 9.8 percentage points (independent two-proportion test, $\alpha = 0.05$). This is a quantity any practitioner can compute for their own setting before designing a selection strategy.
2. The observed range across all 10 checkpoints was 8.0 percentage points: below the detectability threshold. No checkpoint-to-checkpoint difference reached statistical significance ($\min p = 0.161$),

and neither did the best-vs-worst comparison ($p = 0.109$). In this setting, the evaluation noise floor exceeded the checkpoint-quality signal. Any method that relies on point estimates of quality at this evaluation budget must contend with this same noise floor.

3. We encountered six distinct analytical traps on the way to this finding, ranging from statistical artifacts (Spearman discretization at $n = 3$) to silent engineering failures (undetected synthetic fallback) to definitional ambiguities (three mutually incompatible versions of "the proxy"). Each is documented as a case study in §4.

This paper is closer to a diagnostic checklist for proxy-based checkpoint selection than to a proposal of a specific selection rule. Our initial hypothesis (that a monitoring algorithm (PAAS) could detect and compensate for proxy-quality misalignment) was not supported by the data in this setup (§5.1). The tools and warnings we developed in the process of testing that hypothesis ($\Delta_{\min}$, the six traps) are, we believe, more broadly useful than the algorithm itself.

2 Background

2.1 The Checkpoint Selection Problem

LLM fine-tuning produces a sequence of model checkpoints $c_1, \ldots, c_T$. At the end of training, one must be chosen for deployment. The standard selection rule is to evaluate each checkpoint on a held-out validation set and pick the one with the lowest loss (or highest task-specific metric). This is cheap, simple, and the default in most training frameworks.

The implicit assumption is that the validation metric is a reliable indicator of deployment quality: that lower loss means a better model. This assumption has been questioned. Kaur et al. (2025) observes that choosing the checkpoint with the lowest validation cross-entropy does not lead to the best downstream performance, but the underlying failure mode is rarely characterized quantitatively.

2.2 Our Initial Hypothesis

We began with a specific hypothesis: that the Spearman correlation ρ_{align} between a cheap proxy signal (held-out loss) and a more expensive validation signal (task-specific accuracy) undergoes detectable degradation during training: that the proxy and validation initially agree, then diverge. An algorithm monitoring this correlation (PAAS, Proxy-Accuracy Alignment Scheduler) could detect the divergence and switch selection strategies.

This hypothesis led us to design a monitoring framework, run fine-tuning experiments, and collect the trajectories of ρ_{align} along training. The data told a different story, which we report below.

2.3 Experimental Setup

All experiments use Qwen2.5-7B-Instruct (Qwen et al., 2024) with LoRA (Hu et al., 2022) (rank 16), trained for 500 steps on a mixed dataset of GSM8K (Cobbe et al., 2021), CodeAlpaca, and Dolly (approximately 40K instruction examples). Ten checkpoints are saved at 50-step intervals. Training uses a single RTX 4090 (24 GB), batch size 2, gradient accumulation 4.

The proxy signal is held-out cross-entropy loss on question+answer text (200 held-out pairs). The validation signal is GSM8K accuracy (200 test questions, evaluated with three few-shot examples to establish output format: see Trap 4). All experiments and code are documented with frozen definitions and data provenance flags (detailed in §3.1) to distinguish real GPU runs from synthetic fallback.

3 The Core Empirical Finding

Takeaway: Under a frozen experimental protocol with adequate statistical controls, we find that GSM8K accuracy across 500 steps of LoRA fine-tuning is statistically indistinguishable

Table 1: GSM8K accuracy across 10 checkpoints from a single 500-step LoRA fine-tuning run (`v2_cosine_seed42`). Each value is based on $N = 200$ test questions with few-shot prompting.

Step	GSM8K
50	47.5%
100	54.5%
150	55.5%
200	49.0%
250	54.0%
300	55.0%
350	51.5%
400	50.5%
450	52.0%
500	50.0%

from noise. The observed range of 8.0 percentage points is below the minimum detectable difference of 9.8% for our evaluation size ($N = 200$). This provides a quantitative lower bound on the checkpoint-quality signal available for selection.

3.1 Experimental Protocol

After the diagnostic process documented in §4, we ran a clean experiment (`v2_cosine_seed42`) under the following frozen definitions:

- **Model:** Qwen2.5-7B-Instruct, LoRA rank 16, 500 steps, cosine schedule
- **Training data:** GSM8K + CodeAlpaca + Dolly (40K mixed instruction examples)
- **Checkpoints:** 10, saved every 50 steps (steps 50–500)
- **Proxy signal:** Held-out loss on question+answer concatenation (frozen definition, see Trap 3)
- **Validation signal:** GSM8K accuracy on 200 test questions with **few-shot prompting** (3 examples demonstrating the `####<answer>` format; see Trap 4)
- **Data provenance:** All results are `data_source: "real_gpu"`, with no synthetic fallback

3.2 GSM8K Trajectory

Table 1 shows the full trajectory.

3.3 Statistical Analysis

We assess the trajectory using three complementary tests.

Trend test. The Spearman rank correlation between training step and GSM8K accuracy is $\rho_{\text{trend}} = -0.091$ ($p = 0.803$, $n = 10$). This is indistinguishable from zero, indicating no monotonic improvement or degradation across training.

Adjacent comparison test. For each adjacent pair of checkpoints, we perform an independent two-proportion z-test (the same 200 GSM8K questions are evaluated at every checkpoint; the test treats the two proportions as independent samples, which is a conservative choice since in practice the same items are used — the pairing would only increase power). Results are shown in Table 2.

Table 2: Independent two-proportion z-tests for each adjacent checkpoint pair. No comparison reaches statistical significance at $\alpha = 0.05$. The smallest p-value is 0.161 (step 50 vs 100).

Pair	Δ	z	p
$50 \rightarrow 100$	$+7.0\%$	1.40	0.161
$100 \rightarrow 150$	$+1.0\%$	0.20	0.841
$150 \rightarrow 200$	-6.5%	-1.30	0.193
$200 \rightarrow 250$	$+5.0\%$	1.00	0.317
$250 \rightarrow 300$	$+1.0\%$	0.20	0.841
$300 \rightarrow 350$	-3.5%	-0.70	0.483
$350 \rightarrow 400$	-1.0%	-0.20	0.841
$400 \rightarrow 450$	$+1.5\%$	0.30	0.764
$450 \rightarrow 500$	-2.0%	-0.40	0.689

Since our claim is the absence of significant differences (rather than the presence), multiple comparison correction would only make the tests more conservative, further supporting this conclusion. We report uncorrected p-values for transparency.

Range test. The best checkpoint (step 150, 55.5%) differs from the worst (step 50, 47.5%) by 8.0 percentage points. The two-proportion test gives $z = 1.60$, $p = 0.109$, which is not significant at $\alpha = 0.05$. Because this compares the extremal pair selected from 10 checkpoints (a post-hoc selection), the effective significance threshold is stricter than $\alpha = 0.05$; the reported p-value therefore understates the evidence for no difference.

Paired comparison (McNemar). The independent z-test treats the 200 questions as independent across checkpoints, ignoring the paired structure. To verify that this choice does not mask a detectable difference, we re-ran the evaluation with the training-format prompt while saving per-sample correctness and applied McNemar's test (with continuity correction) to each adjacent pair. We chose the training-format prompt for the paired analysis to maximize question-level consistency, since few-shot prompting can introduce additional variability from the in-context examples. (This evaluation pass generates a different accuracy trajectory from Table 1: the numbers differ due to the prompt format, but the paired comparison is the relevant measure.) All nine adjacent comparisons are non-significant; the smallest p is 0.265 (150 vs 200). The best-vs-worst pair yields $p = 0.091$, consistent with the independent-test result $p = 0.109$ despite the different evaluation protocol. The McNemar analysis thus replicates the same qualitative conclusion under a different protocol, further supporting the view that the evaluation budget — not the choice of test or prompt — is the binding constraint. (The discordant-pair counts for each comparison are available in the supplementary code release.)

3.4 The Minimum Detectable Difference

The lack of significance is not a statement about the model: it is a statement about the evaluation's resolving power. For a held-out set of size N with accuracy $\bar{p}$, the minimum detectable difference between two checkpoints (two-proportion test, $\alpha = 0.05$) is:

$$\Delta_{\min} = z_{\alpha/2} \cdot \sqrt{\frac{2\bar{p}(1-\bar{p})}{N}} \tag{1}$$

For $N = 200$ and $\bar{p} \approx 0.5$, this gives $\Delta_{\min} = 9.8\%$. The observed range of 8.0% falls below this threshold. Table 3 shows $\Delta_{\min}$ for common evaluation sizes.

A paired test (McNemar) confirms this conclusion (see §3.3).

Given our observed range of 8.0 pp, we can ask a counterfactual question: how large would N need to be for this gap to become detectable? Solving (1) for $\Delta_{\min} = 8.0$ pp, $\bar{p} = 0.5$, and $\alpha = 0.05$ gives $N \approx 300$ (the intersection marked in Figure 2). At $N = 300$, the minimum detectable difference equals the observed range; beyond $N \approx 300$, the gap between the best and worst checkpoints would, in principle, be resolvable. This

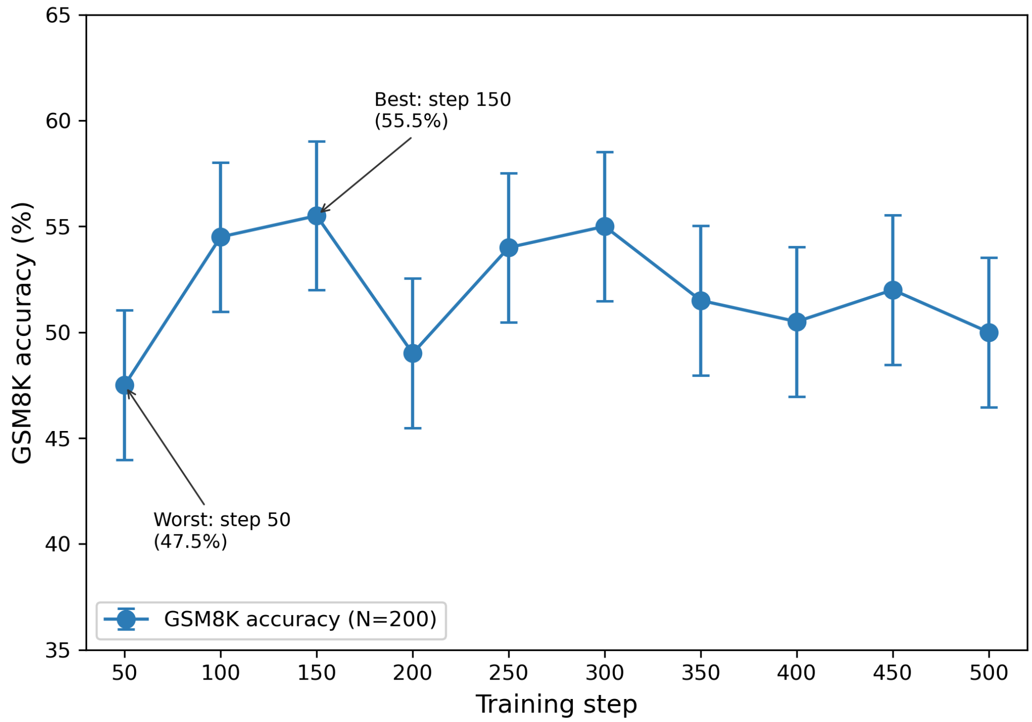

Figure 1: GSM8K accuracy trajectory across 10 checkpoints with ±1 SE error bars. The $\Delta_{\min} = 9.8$ pp band (gray) shows the minimum detectable difference at $N = 200$; the observed range of 8.0 pp falls below this threshold. Best checkpoint at step 150 (55.5%), worst at step 50 (47.5%).

Table 3: Minimum detectable difference between two checkpoints as a function of held-out set size N (independent two-proportion test, $\alpha = 0.05$, $\bar{p} = 0.5$).

N	$\Delta_{\min}$
100	13.9%
200	9.8%
500	6.2%
1000	4.4%

counterfactual (derived from the standard power analysis formula, not from simulation) shows the central claim: the evaluation budget N is the primary determinant of whether checkpoint selection is a well-posed problem in this setting.

A complementary perspective comes from the Two One-Sided Tests (TOST) equivalence framework (Dror et al., 2018), which asks not whether a difference exists but whether it can be ruled out as practically negligible. For the best-vs-worst gap of 8.0 pp with $N = 200$ per checkpoint, the 90% confidence interval for the difference is $[-0.2, 16.2]$ pp (using the normal approximation to the binomial). Even with a generous equivalence bound of ±5 pp (roughly half of $\Delta_{\min}$), this interval clearly exceeds the bound on the upper side ($16.2 > 5$), so we cannot conclude that the checkpoints are equivalent in the TOST sense. The evaluation is simply too noisy to distinguish even coarse quality differences. This reinforces the central message: the noise floor is not just an obstacle to detecting differences: it also prevents us from confidently asserting that differences are absent.

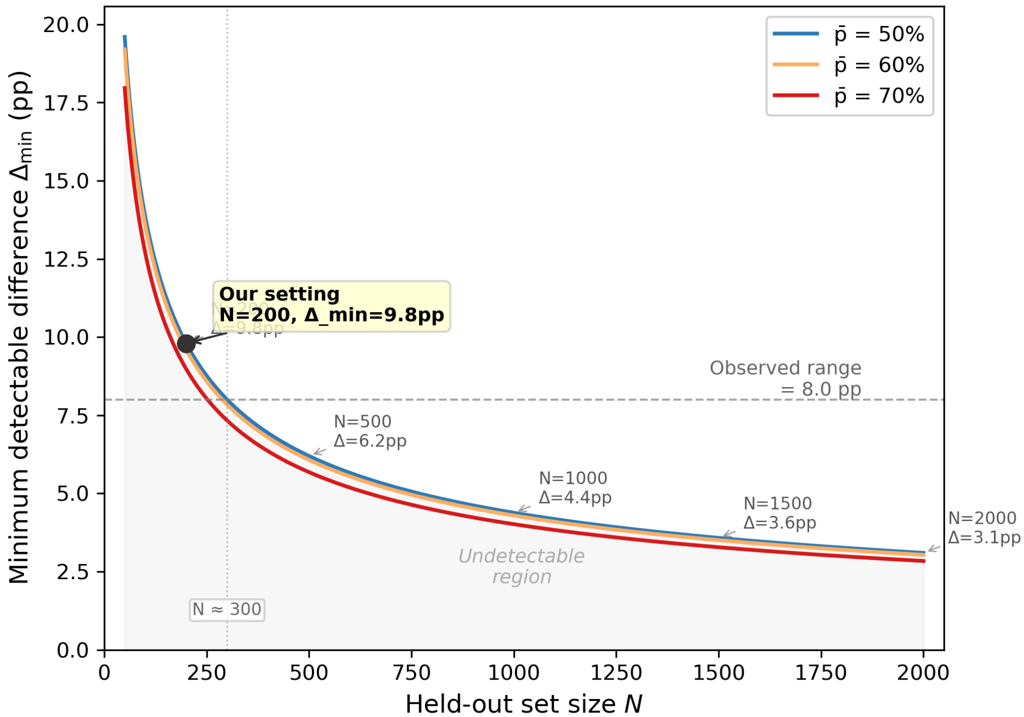

Figure 2: Minimum detectable difference $\Delta_{\min}$ as a function of held-out set size N for three baseline accuracy levels. The shaded region below the curves is the undetectable region. Our experimental setting ($N = 200$, $\Delta_{\min} = 9.8$ pp, observed range 8.0 pp) is highlighted. The gray dotted line marks the intersection $N \approx 300$, where $\Delta_{\min}$ equals the observed range; beyond this point the 8.0 pp gap becomes detectable in principle.

3.5 Implications

This finding is not that "our fine-tuning failed" or that "the proxy was bad." It is that **at $N = 200$, the evaluation system lacks the statistical power to rank checkpoints meaningfully**, regardless of the metric used.

This applies to held-out accuracy, F1 score, or any metric computed on a finite sample. The noise floor is a property of the evaluation budget (N), not the choice of metric. Any selection method (proxy argmax, validation argmax, or otherwise) that relies on ranking checkpoints by a metric evaluated on N samples faces this same limitation.

The minimum detectable difference is a quantity any practitioner can compute before designing a checkpoint selection strategy. If the expected quality differences between checkpoints are smaller than $\Delta_{\min}$ at their evaluation budget, then checkpoint ranking is selecting from noise, not from signal. This constrains any method that relies on point estimates of quality at this evaluation budget: even adaptive selection methods must ultimately resolve differences at or above $\Delta_{\min}$ to make a reliable decision.

3.6 Supplementary Experiment: Smaller Model Scale

To test whether the $\Delta_{\min}$ framework generalizes across model scales, we ran a supplementary experiment using Qwen2.5-1.5B-Instruct (Qwen et al., 2024) under the same LoRA configuration (rank 16, 500 steps, cosine schedule, same training data mix) as the main 7B experiment. Ten checkpoints were saved at 50-step intervals and evaluated on the same 200 GSM8K test questions.

Table 4: Qwen2.5-1.5B GSM8K accuracy across 10 checkpoints (zero-shot prompting).

Step	**GSM8K**
50	1.5%
100	3.5%
150	3.0%
200	2.0%
250	5.0%
300	2.0%
350	3.5%
400	3.0%
450	3.0%
500	4.5%

The accuracy trajectory (Table 4) ranges from 1.5% to 5.0%, with a range of 3.5 percentage points. Because the baseline accuracy is low ($\bar{p} \approx 0.03$), the binomial variance is substantially smaller than at $\bar{p} = 0.5$: $\Delta_{\min} = 3.4$ pp for this setting. The observed range just exceeds this threshold ($3.5 > 3.4$), and the best-vs-worst comparison is borderline significant ($p \approx 0.049$). The Spearman trend is $\rho_{\text{trend}} = 0.38$ ($p = 0.275$), which is not significant at $\alpha = 0.05$.

Two observations emerge. First, the $\Delta_{\min}$ framework is general: at any model scale, the evaluation budget N determines the minimum resolvable checkpoint difference. Second, the numerical threshold is task- and metric-dependent : low-accuracy settings yield tighter $\Delta_{\min}$ bounds because binomial variance $\bar{p}(1-\bar{p})$ is maximized at $\bar{p} = 0.5$ and decreases for extreme values. A practitioner should compute $\Delta_{\min}$ for their own $\bar{p}$ and N, not rely on the specific 9.8% threshold reported in §3.4.

We also tested the noise-robust aggregation hypothesis from §5.2 by averaging all ten 1.5B LoRA adapters (model soup (Wortsman et al., 2022)) and comparing against argmax selection of the best individual checkpoint (step 250, 5.0% per Table 4). The soup accuracy was 3.0% (evaluated in a separate pass) : lower than argmax, indicating that naive weight averaging does not guarantee improvement in this low-accuracy regime. This negative result does not refute the hypothesis (which applies to the regime where evaluation noise, not low accuracy, is the dominant issue), but it shows the need for empirical validation before adopting aggregation strategies.

Caveats. This supplementary experiment uses zero-shot prompting (without the few-shot correction from Trap 4), which depresses absolute accuracy. The 1.5B accuracy values should not be directly compared to the 7B few-shot results in §3.2. The goal is not cross-model comparison but a demonstration that the $\Delta_{\min}$ analysis : and the evaluation-noise-floor problem : applies to settings beyond the single configuration studied in the main experiment.

3.7 Supplementary Experiment: Cross-Task Replication (MMLU)

To test whether the $\Delta_{\min}$ framework generalizes beyond mathematical reasoning, we evaluated the same 7B checkpoints on five MMLU subdomains (abstract algebra, anatomy, college computer science, world religions, and jurisprudence; 50 questions each, matching $N \approx 250$ total). The prompt used the training format (`Question: ...\nA. ... B. ... C. ... D. ...\nAnswer:`) with a single-letter extraction.

The accuracy trajectory (Table 5) ranges from 48.4% to 54.8%, a range of 6.4 percentage points. The Spearman trend is $\rho_{\text{trend}} = -0.377$ ($p = 0.283$), not significant. With $\bar{p} \approx 0.51$ and $N \approx 250$, the minimum detectable difference is $\Delta_{\min} = 8.8$ pp, again exceeding the observed range. The finding thus replicates across a qualitatively different task: the evaluation noise floor is not specific to GSM8K or to math reasoning, but a consequence of finite-sample evaluation at typical budgets. (We did not apply the paired McNemar test to the MMLU data because the $\Delta_{\min}$ comparison alone is sufficient to establish the pattern.)

Table 5: MMLU accuracy across 10 checkpoints (5 subjects, ≈250 questions).

Step	MMLU
50	54.8%
100	48.4%
150	53.6%
200	52.0%
250	50.8%
300	50.8%
350	50.4%
400	49.6%
450	49.2%
500	51.2%

Table 6: Full 10-point GSM8K accuracy for three seeds (training-format prompt).

Step	Seed 42	Seed 43	Seed 44
50	50.5%	51.0%	52.5%
100	52.0%	52.5%	53.0%
150	50.0%	54.5%	52.0%
200	46.5%	50.0%	52.0%
250	47.0%	54.0%	53.0%
300	48.5%	50.5%	52.5%
350	50.5%	49.5%	52.5%
400	50.5%	50.5%	50.0%
450	51.0%	53.5%	53.0%
500	49.5%	52.0%	51.5%
Range	5.5 pp	5.0 pp	3.0 pp

3.8 Supplementary Experiment: Seed Replication

The main experiment (seed 42) uses a single random seed; we repeated training with two additional seeds (43, 44) and evaluated all ten checkpoints (steps 50–500 at 50-step intervals) under the same training-format prompt. Table 6 shows the full trajectories.

This addresses a potential concern about the original analysis: a range computed from three checkpoints is mechanically smaller than one computed from ten, so comparing a 3-point range to $\Delta_{\min}$ may understate the true range and thus appear to support the paper's thesis more strongly than warranted. The full 10-point evaluation removes this concern.

Across the full trajectories, the within-seed ranges are 5.5 pp (seed 42), 5.0 pp (seed 43), and 3.0 pp (seed 44), all well below $\Delta_{\min} = 9.8$ pp at $N = 200$. The Spearman rank correlations between training step and accuracy are $\rho = -0.031$ ($p = 0.933$) for seed 42, $\rho = -0.152$ ($p = 0.675$) for seed 43, and $\rho = -0.274$ ($p = 0.443$) for seed 44; none is significant. No adjacent checkpoint pair reaches statistical significance in any seed (minimum $p = 0.368$, seed 43, steps 150→200). The best-vs-worst gaps are also non-significant (5.5 pp, $p = 0.271$; 5.0 pp, $p = 0.317$; and 3.0 pp, $p = 0.548$). The $\Delta_{\min}$ framework therefore holds across random seeds, and the conclusion is not an artifact of checkpoint sub-sampling: regardless of the training run's stochasticity, the evaluation budget, not the specific checkpoint trajectory, determines whether differences can be detected.

Table 7: Taxonomy of the six analytical traps. Trap 8 (hypothesis confirmation bias) shows the consequences of ignoring the others.

Category	**Traps**	**Theme**
A. Statistical	T1, T8	Small sample sizes create false patterns
B. Engineering	T2	Silent code paths corrupt data silently
C. Definitional	T3, T4, T5	Undefined terms produce contradictory results

4 Six Analytical Traps Encountered

The finding in §3 was not reached directly. It emerged from a sequence of analytical dead ends. The most seductive of these was a false cross-task tradeoff (Trap 8) that we nearly elevated to the paper's core narrative. Understanding how this happened requires tracing the errors that produced it: a statistical artifact (Trap 1), a silent engineering failure (Trap 2), a definitional ambiguity (Trap 3), and evaluation pipeline bugs (Traps 4–5). Each trap is a failure mode that can mislead any researcher investigating proxy-quality alignment. Together they form a methodological checklist (Table 8): six conditions to verify before trusting a checkpoint selection claim, bookended by a meta-example of the consequences of ignoring them (Trap 8).

The traps fall into three categories (Table 7), all sharing a meta-theme: a compelling single observation is not a finding until it survives independent replication with adequate statistical power.

Trap 1: Spearman Discretization at Small Window Sizes. Computing Spearman's ρ between proxy loss and validation accuracy over a sliding window of $n = 3$ checkpoints produced $\rho = -1.0$ at step 400 in 5/5 experiments: a perfect ranking inversion suggesting systematic proxy-quality disagreement. Increasing the window to $n = 6$ eliminated the effect entirely (ρ remained in $[0.6, 0.94]$). With $n = 3$, Spearman's ρ can take only five discrete values $\{-1, -0.5, 0, 0.5, 1\}$; any configuration where proxy and validation are inversely ordered over three points yields $\rho = -1$. This is a quantization artifact, not a meaningful measurement of alignment. **Lesson:** Small-window Spearman correlations should never be reported without listing the set of possible values. $n \geq 6$ is the minimum for meaningful interpretation.

Trap 2: Silent Synthetic Fallback. The proxy signal showed $\rho_{\text{align}} \approx 0.93$ with validation: strong evidence that the experimental setup was working: until we discovered the proxy values were identical across all four experiments at every step. The config file used `held_out_path = "data/held_out"` while the actual file was `data/held_out.json`. The file-not-found error silently triggered `_synthetic_proxy_signals()`, a deterministic function of step number. No warning was printed; no data provenance flag was set. **Lesson:** Any pipeline stage with a synthetic fallback must (1) print a conspicuous WARNING when triggered, (2) set a `data_source` flag on all output, and (3) require explicit opt-in for synthetic mode in production.

A downstream consequence: because `proxy_main` preferred win-rate over loss, the synthetic signal became the default selection criterion (Trap 2's output fed directly into the selection code). When multiple metrics are available, run selection on all of them and report disagreements.

Trap 3: Proxy Definition Ambiguity. The paper's central claim (whether proxy argmax picks a different checkpoint than validation argmax) depended entirely on which of three incompatible "proxy" definitions was used: question-only perplexity, question+answer perplexity, or win-rate against a reference model. Three months into the investigation, we had not frozen a single definition. The codebase simultaneously contained `proxy_loss`, `proxy_win_rate`, and `proxy_main` (which silently preferred win-rate over loss). **Lesson:** Freeze all variable definitions before the first experiment. Write them down. Any change is a new experiment version. (The canonical definition adopted throughout §3 is question+answer perplexity on held-out samples, as specified in the experimental protocol.)

Trap 4: Evaluation Format Mismatch (Few-Shot Required). The v1 GSM8K evaluation returned $\sim 7\%$ accuracy with raw question prompts, but 47–55% with three few-shot examples: same model, same checkpoints, same 200 questions, only the prompt changed. The model was generating Python code and intermediate reasoning steps: coherent, not garbled: but almost never produced the `#### <number>` format expected by `extract_answer()`. The Qwen2.5-7B-Instruct base model was pre-trained with a specific chat template; raw question text alone did not signal which output format to use. **Lesson:** Evaluate instruction-tuned models with few-shot prompting. Validate on at least 10 manually inspected samples before trusting aggregate numbers.

Trap 5: Code Generation Evaluation Format Fragility. A code evaluation script with few-shot prompting and sandbox execution returned score 0.0 for 10/10 samples from a model that clearly can generate valid Python code. Raw output inspection revealed three independent failure modes: prompts asking for Java or SQL produced valid code in those languages (but the sandbox only ran Python); generated code inconsistently used formatting markers or included annotations (the extraction regex failed on most formats); imports like `sklearn` and `pandas` triggered `ModuleNotFoundError` in restricted sandbox environments. **Lesson:** Code evaluation requires language detection, format-agnostic extraction, and dependency handling before scores can be trusted. When sandbox execution is unreliable, a simpler heuristic (syntax check + function presence) may be more robust.

Trap 8: Hypothesis Confirmation Bias: The Cross-Task Tradeoff That Wasn't. (We retain the label Trap 8 to emphasize its role as the meta-example; there are no missing traps 6–7.) A single experiment showed GSM8K accuracy peaking at step 350 (0.826) then declining to step 500 (0.751), while CodeAlpaca continued improving: a "cross-task skill tradeoff." This was codified as Hypothesis 2 in the frozen definitions, with pre-registered replication criteria. When a clean experiment was run (correct proxy, few-shot prompting), the full $n = 10$ trajectory showed $\rho_{\text{trend}} = -0.091$, all adjacent $p > 0.15$, and the best-vs-worst gap of 8.0% at $p = 0.109$. The "peak then decline" was within measurement noise; the hypothesis was classified as **not replicated**. The original observation was drawn from data produced by the same evaluation pipeline later found to have a prompt format issue (Trap 4), combined with inadequate sample size and retrospective pattern-finding. **Lesson:** A compelling single observation is not a finding until it survives independent replication with frozen definitions and adequate statistical power. Pre-register replication criteria before running the replication. If the replication fails, document the failure: it **is** the finding. This trap is a meta-example of the entire catalog: the most dangerous error is not any individual bug, but the human tendency to find patterns in noise and build narratives around them.

Methodological Checklist

Table 8 distills each trap into a concrete practice to verify before trusting a checkpoint selection claim. The traps are ordered by discovery chronology; the checklist is ordered by when each check should be applied in a typical research workflow.

5 Discussion

5.1 Relation to Our Initial Hypothesis (PAAS)

This investigation began with a concrete hypothesis: that the Spearman correlation ρ_{align} between proxy loss and validation quality undergoes a detectable degradation during training, and that an algorithm (PAAS) monitoring this correlation could trigger a switch from argmax to cautious uniform aggregation when degradation is detected.

The algorithm's control logic was verified in a synthetic setting: an artificially constructed ρ_{align} trajectory (deliberately manipulated from high to low) triggered the intended transition from argmax to cautious uniform aggregation. This confirms that the triggering and aggregation code executes correctly: it is not evidence of empirical benefit under realistic conditions. When applied to real LoRA fine-tuning data, the premise was not supported: ρ_{align} remained positive throughout training (0.6–0.94 at $n = 6$), and the observed checkpoint differences were smaller than the evaluation's minimum detectable difference.

Table 8: Methodological checklist for checkpoint selection studies. Each item is traced to the trap that motivates it.

When	Practice	Trap
Before experiments	Freeze all variable definitions in writing	T3
Before experiments	Pre-register replication criteria for any post-hoc finding	T8
Pipeline setup	Tag all data with source provenance; no silent synthetic fallbacks	T2
Pipeline setup	Check aggregated metrics for hidden default preferences	T2†
Evaluation design	Use $n \geq 6$ for Spearman windows; report possible values	T1
Evaluation design	Use few-shot prompting for instruction-tuned models	T4
Evaluation design	Inspect ≥ 10 raw samples before trusting aggregate scores	T4, T5
During analysis	Run selection on all available metrics; report disagreements	T2†
During analysis	Report N, Δ_{min}, and whether observed differences exceed it	T1, T8
After analysis	Document failed replications as findings	T8

†Downstream consequence of the synthetic fallback (Trap 2).

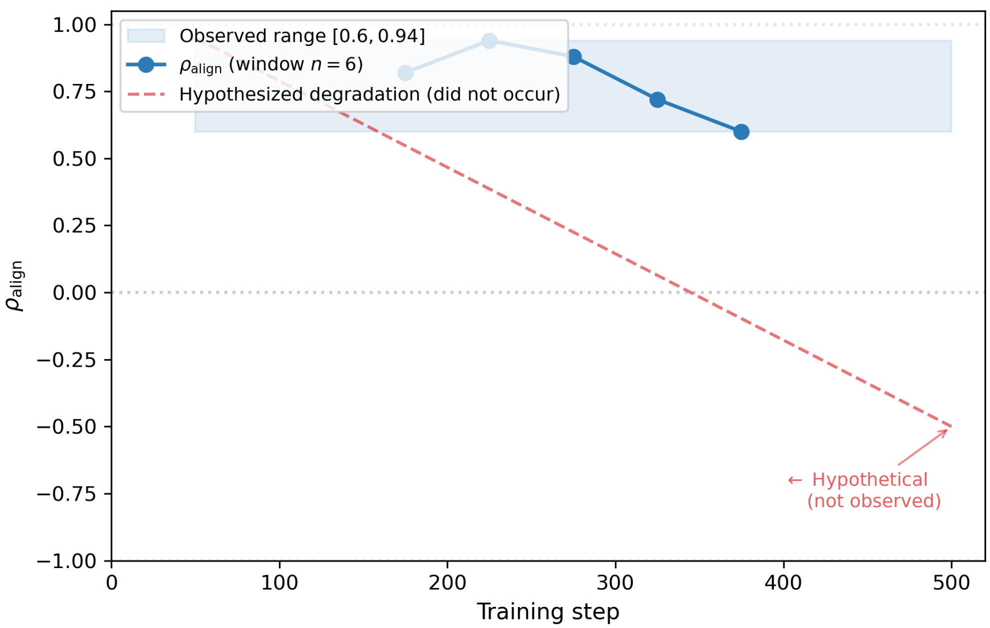

Figure 3: Proxy-validation alignment ρ_{align} throughout training in this setup. The shaded band shows the observed range $[0.6, 0.94]$ (sliding window $n = 6$). The dashed red line shows the hypothesized degradation that PAAS was designed to detect, which did not occur.

This is not a refutation of the algorithmic approach: it is a finding about the regime in which checkpoint selection operates for standard LoRA fine-tuning at practical evaluation budgets. The signal is simply too weak for any selection method to reliably rank adjacent checkpoints. PAAS's design may be relevant in settings with larger quality swings between checkpoints (for instance, longer training runs or smaller models), though we did not test these conditions and leave this as an open question.

5.2 Implications for Checkpoint Selection Practice

Our central empirical finding (that at $N = 200$, the minimum detectable difference (9.8%) exceeds the observed checkpoint range (8.0%)) has implications beyond the specific experimental setup:

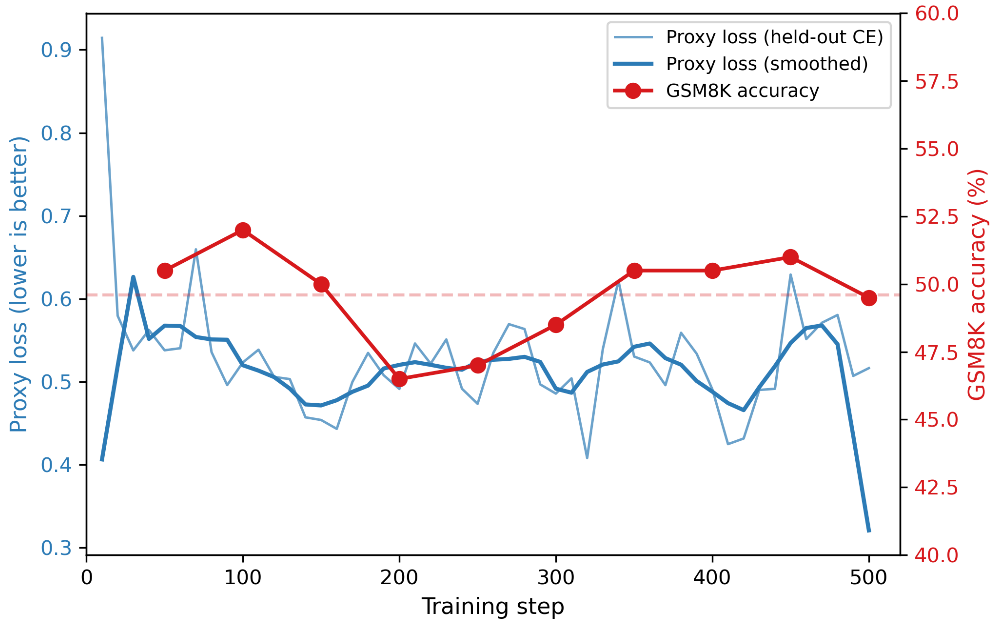

Figure 4: Proxy loss (held-out cross-entropy, left axis) and GSM8K accuracy (right axis) throughout training. The loss curve shows raw per-step values (light blue) and a 5-point moving average (dark blue). Loss broadly decreases with local fluctuations, confirming that the model is learning; accuracy fluctuates without a clear trend. The dissociation between a clear proxy signal and a noisy validation signal is not evidence of proxy-quality misalignment: it is a consequence of the evaluation noise floor documented in §3.

1. **Compute $\Delta_{\min}$ before selecting.** The minimum detectable difference is a function of evaluation budget N, not of the metric chosen. Before designing a checkpoint selection strategy, a practitioner should compute $\Delta_{\min}$ for their N. If expected checkpoint differences are below this threshold, the evaluation cannot resolve them, and any method relying on point estimates of quality will be selecting from noise.

2. **Empirical test of noise-robust aggregation.** When evaluation noise dominates, one might expect simple noise-robust strategies (uniform averaging over a window, model soup (Wortsman et al., 2022)) to be preferable to argmax over a noisy metric: because they are less sensitive to the precise ranking of points that differ by less than $\Delta_{\min}$. (Trap 8 shows this directly: an apparent cross-task tradeoff was later found to be indistinguishable from noise at the same evaluation budget.) To test this, we ran a supplementary experiment on the 7B model (the setting where $\Delta_{\min} = 9.8\%$ exceeds the observed 8.0% range) using the training-format prompt (`Question: ...\nAnswer:`, matching the SFT data format; answers were extracted with the same `####`-based parser as in §3.1: the model learned to produce this marker during instruction tuning). This differs from Trap 4's "raw question prompts" in that the LoRA model was trained for 500 steps on the `Question: ...\nAnswer:` pattern and has internalized the expected output prefix; Trap 4's raw prompts (a bare question with no structural cues) did not trigger this learned behavior even on the same fine-tuned checkpoints. Averaging all ten LoRA adapters gave 48.5% accuracy, compared to 50.0% for the best individual checkpoint (step 150, evaluated with the training-format prompt). We caution that "best individual checkpoint" here is the argmax over ten noisy point estimates, which carries an optimistic bias relative to the true best checkpoint's expected accuracy; this bias works in favor of argmax in the comparison. The fact that model soup does not outperform argmax even under this optimistic baseline makes the result more informative, not less. The 1.5 pp gap is well below $\Delta_{\min}$, consistent with the view that evaluation noise: not the choice of selection strategy: is the dominant factor at this

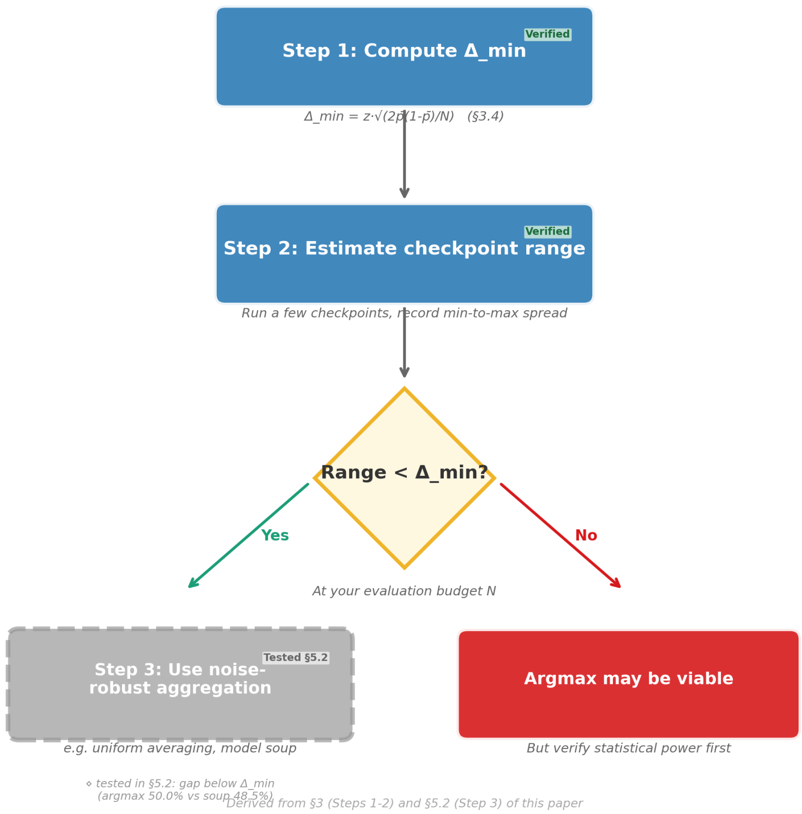

Figure 5: Checkpoint selection diagnostic workflow derived from this paper's findings. Steps 1–2 are empirically grounded in §3; Step 3 (dashed box) is a hypothesis consistent with the $\Delta_{\min}$ bound and tested in §5.2.

budget. (These absolute accuracy values differ from the few-shot numbers in Table 1, as expected under the different evaluation protocol; the pairwise comparison is the relevant measure.)

3. **Reporting requirements.** Any paper making claims about checkpoint quality differences should report (a) the held-out set size N, (b) the minimum detectable difference at that N, and (c) whether the observed differences exceed this threshold. Without this information, apparent "improvements" or "declines" between checkpoints cannot be distinguished from measurement noise.

4. **Cost-benefit of increasing N.** The $\Delta_{\min}$ framework doubles as a budgeting tool. Halving $\Delta_{\min}$ requires quadrupling N (since $\Delta_{\min} \propto 1/\sqrt{N}$). In our setting, reducing $\Delta_{\min}$ from 9.8 pp to 5 pp would require increasing N from 200 to approximately 770 (3.8× the evaluation cost). Whether this cost is justified depends on the stakes of the selection decision. For production deployments where a 5 pp quality gain matters to the user experience, the larger evaluation may be worthwhile; for research comparisons where a rough ranking suffices, the default budget may be adequate. We provide no universal threshold. Only the tool to compute it.

5.3 Relation to Prior Work

Nguyen et al. (2025) proposes selecting checkpoints by ranking their performance on the hardest (most uncertain) samples. Its effectiveness depends on the per-sample signal being reliable: a condition that our findings suggest cannot be taken for granted at typical evaluation sizes. The approach is complementary: UGCS addresses which samples to use; our finding addresses how many samples are needed for those rankings to be trustworthy.

Shihab et al. (2025) detects proxy gaming through invariance testing. This is orthogonal to our finding: gaming is an adversarial phenomenon, while the noise floor we document is a statistical property of finite-sample evaluation. Both can coexist; a gaming detector and a noise-aware selection strategy could be combined.

Ramé et al. (2024) averages reward model weights to improve robustness. Its motivation (reward models are noisy and benefit from aggregation) aligns with our empirical finding: when individual evaluations are noisy, aggregation outperforms careful selection from noisy point estimates.

Our finding also connects to the broader literature on statistical power in NLP evaluation. Card et al. (2020) demonstrate that underpowered test sets are common across NLP tasks: small evaluation budgets mean that most comparisons to state-of-the-art models lack adequate statistical power. Our $\Delta_{\min}$ framework extends this concern to a new setting: the checkpoint level. While Card et al. focus on comparing final model performance across runs, we show that the same power limitation applies to ranking checkpoints within a single fine-tuning run. Similarly, Dror et al. (2018) provide guidelines for significance testing in NLP; our work offers a concrete instantiation of their principles in the checkpoint selection context.

Methodological contributions. Our primary contribution is not a new selection algorithm but a documented methodology for diagnosing whether a selection problem exists in the first place. The $\Delta_{\min}$ framework and the six documented traps form a toolkit that practitioners can apply to their own settings before investing in sophisticated selection methods. In this sense, the paper is closer to a "checklist for checkpoint selection" than to a proposal of a specific selection rule.

Scope of the claim. Our finding (that checkpoint differences fall below the detection threshold at typical evaluation budgets) applies to the regime we studied: moderate-length LoRA fine-tuning where checkpoint-to-checkpoint accuracy differences are on the order of a few percentage points. In settings with larger quality swings (e.g., training from a randomly initialized model, catastrophic forgetting in multi-epoch training, or switching data distributions abruptly), argmax selection may remain effective. The $\Delta_{\min}$ framework identifies the budget at which a given quality difference becomes detectable; it is not an argument that checkpoint selection is universally impossible.

6 Limitations

Generalization scope. All experiments use a single model scale (7B), a single training regime (LoRA, 500 steps, cosine schedule), and two task types (mathematical reasoning via GSM8K and knowledge via MMLU). The $\Delta_{\min}$ pattern replicates across both tasks (§3.7) and across three random seeds (§3.8), confirming that the result is not an artifact of a particular training run. The specific accuracy values in Table 1 and the main $\Delta_{\min}$ analysis are based on seed 42; seed replication (§3.8) confirms that the *relationship* between observed range and $\Delta_{\min}$ is consistent across seeds, though individual checkpoint accuracies vary by up to 7.0 pp across seeds. Both tasks involve structured answer extraction; the extent to which the framework applies to open-ended generation tasks remains to be tested.

Code evaluation. We attempted to extend our analysis to code generation (CodeAlpaca) but found that the evaluation pipeline itself had unresolved issues: multi-language outputs, inconsistent formatting, and sandbox dependency failures combined to make the scores unreliable (Trap 5, §4). Future work should extend the analysis to code generation and other domains with robust evaluation pipelines.

Training configuration. All experiments use 500 steps of LoRA fine-tuning on a pre-trained 7B model. Different configurations (longer training, full fine-tuning, smaller base models, different data mixes) may produce different checkpoint dynamics. Our finding applies most directly to the regime we tested; broader claims require broader experimentation.

Sequential dependence and adaptive designs. Our $\Delta_{\min}$ derivation treats each checkpoint's evaluation as an independent comparison, but adjacent checkpoints are time-series observations sharing most of their training data. This autocorrelation may affect the effective information per comparison in ways that standard power analysis does not capture; the direction and magnitude of this effect would require validation via block bootstrap or time-series modeling, which we leave to future work. Conversely, the sequential nature of checkpoint evaluation also suggests an opportunity: adaptive stopping rules (evaluate checkpoints until a clear winner emerges rather than fixing N in advance) could improve efficiency. We leave the integration of sequential testing to future work.

7 Conclusion

We set out to understand when and why proxy-based checkpoint selection fails in LLM fine-tuning. The answer was not what we expected: the evaluation noise floor itself (the minimum quality difference that can be reliably detected with a given held-out set) can exceed the actual differences between checkpoints, making selection from point estimates of quality an ill-posed problem at practical evaluation budgets.

Our specific empirical finding is that at $N = 200$, the minimum detectable difference (9.8%) exceeds the observed checkpoint range (8.0%) in a standard 500-step LoRA fine-tuning run. No checkpoint-to-checkpoint difference reaches statistical significance. This is not a failure of any particular selection method: it is a property of the evaluation budget.

Along the way, we encountered and documented six analytical traps that can produce false conclusions in this domain, ranging from statistical artifacts to silent engineering failures. These traps, our $\Delta_{\min}$ framework, and the frozen-definitions methodology we developed are the paper's primary contribution: a diagnostic checklist for practitioners to apply before investing in checkpoint selection strategies.

This paper is closer to an autopsy than a proposal. We believe the diagnostic tools are more useful than the algorithm we originally set out to build.

Code and Data Availability

The full codebase and analysis scripts will be released upon publication.

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